3rd place
PlantCLEF 2026 · private F1 0.439

Multi-Scale ViT Inference with Habitat-Fit Priors and kNN Retrieval for Multi-Species Plant Identification

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The task

train on single plants, predict every species in a quadrat



Train: **1.4M** Pl@ntNet photos · **7,806** species · one plant per image



Test: **2,105** quadrats · 0.5 × 0.5 m · ≈3000 × 3000 px · many species each

Metric: per-image F1, averaged per transect. · Only **11%** of the test set is visible on the public leaderboard.

Pipeline at a glance

one classifier, five corrections, all on cached per-tile probabilities

Tile
3×3 + 4×4 + 5×5 + 6×6
86 tiles / image
+0.016

→

ViT-L/14
DINOv2, fine-tuned by us on 1.4M images
softmax over 7,806
+0.036

→

⊕ kNN
20 neighbours in a FAISS index of 860k ArcFace embeddings
+0.013

→

Max-pool
strongest evidence per species across all tiles

→

Temporal fusion
blend with repeat visits of the same plot
+0.004

→

Habitat-fit
demote species far from the site or at the wrong altitude
+0.041

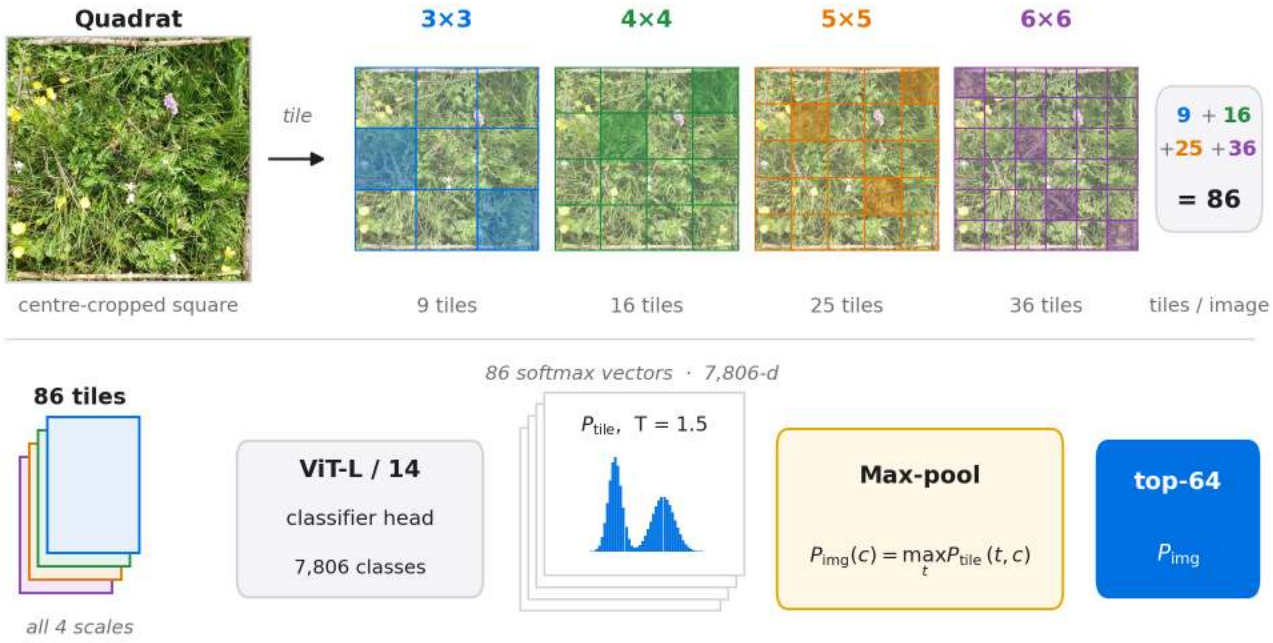
→

Mask + admit
SW-Europe mask threshold · top-10 · floor

Deltas are private-F1 contributions from the cumulative ablation. Full pipeline: **0.458** private F1 (2025 DS@GT entry: 0.348).

Multi-scale tiling

86 tiles, max-pooled



Max, not mean

A species that fills one tile is really there. Averaging over 86 tiles would bury it. Max is also indifferent to tile count, so four grids mix safely.

3+4+5+6 is the sweet spot

Best on the private set for *both* ViT-B and ViT-L. Adding 7×7 and 8×8 raised the public score and lowered the private one.

kNN retrieval

a second opinion from the same backbone



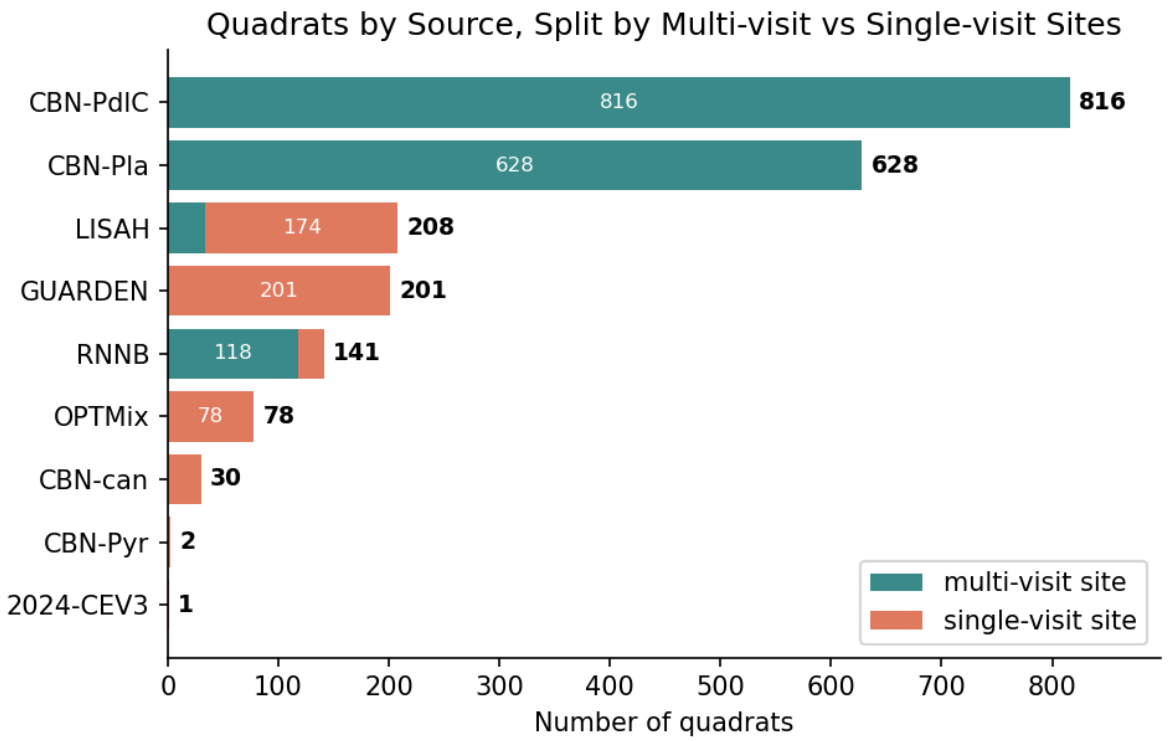
$$P_{tile} = \beta \cdot softmax(z / T) + (1 - \beta) \cdot P_{kNN}$$

k = 20 **τ = 0.07** **β = 0.70 on multi-visit quadrats** **kNN off on single-visit: it cost up to 0.08 F1 there**

Embeddings: LoRA (rank 16) + ArcFace head trained on the frozen ViT-L · 1024-d · 860k training images of South-Western European species

Temporal fusion

the test set is mostly repeat visits



Siblings from the ID alone

RNNB–1–7–**20230512** → strip the date → same plot, other visits.

76% of quadrats have at least one sibling.

Blend

$$P_{fused} = \alpha \cdot P_{img} + (1 - \alpha) \cdot P_{siblings}$$

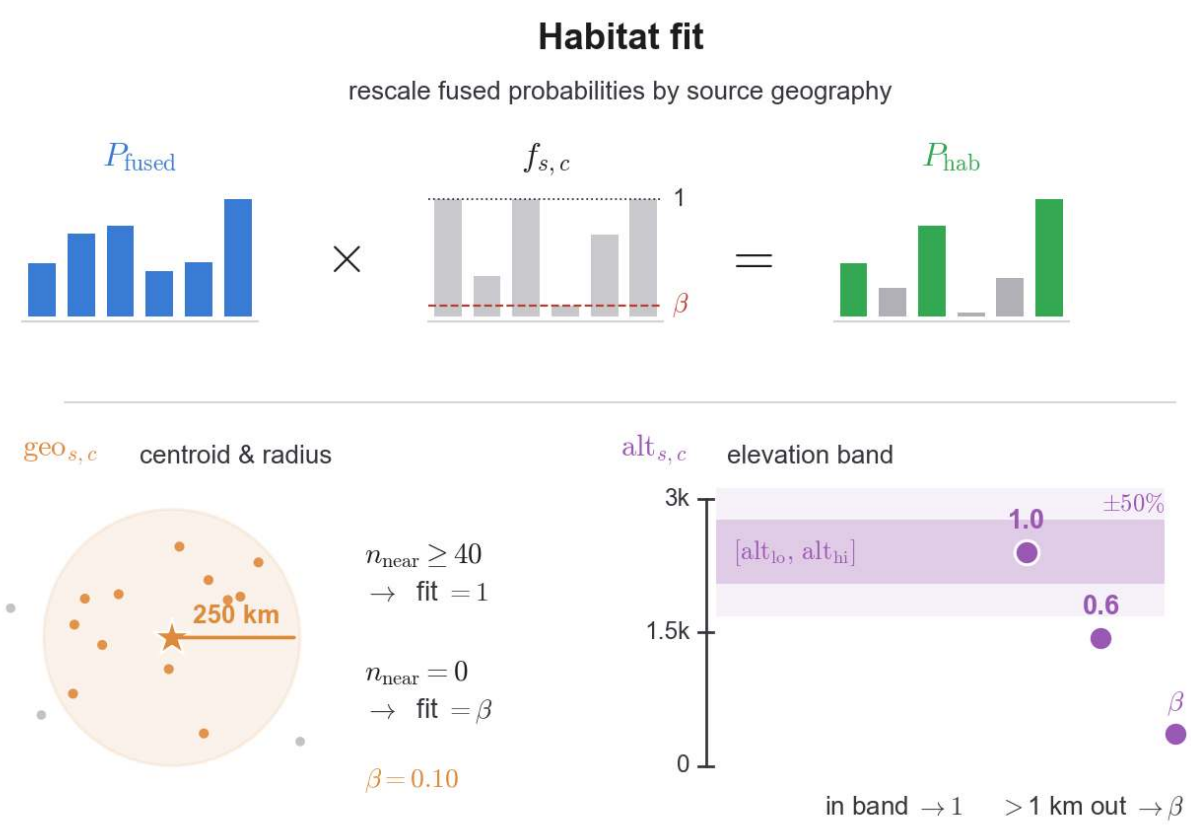
α = 0.15 default · 0.30 for RNNB

$P_{siblings}$ = max over siblings; similarity-weighted for RNNB (coastal dune, strong seasons)

Uses only ID strings and test-tile embeddings. No labels. Smallest contributor (+0.004) but consistent.

Habitat-fit demotion

largest single contributor: +0.041 F1



- The classifier never sees **where** a photo was taken. Every training photo has **GPS + altitude**.
- Per source: a data-derived **centroid** and an **altitude band**.
- **Geo:** ≥ 40 training photos within 250 km → 1. None → 0.1.
- **Altitude:** inside the band → 1. More than 1 km out → 0.1.
- Multiply. It only ever **demotes**.

Then: drop ≈3,000 species centred outside FR/ES/IT/CH; keep P ≥ 0.085, cap at 10, floor of 3 for under-predicting sources.

Results

what we selected vs. what was best

Submitted run	Private	Public
Single pipeline (RNNB β = 0.55)	**0.43902**	**0.51096**
Single pipeline (OPTMix k = 5)	0.39844	0.49842
Ensemble: per-source RRF	0.43553	0.52739
Ensemble: interleaved	0.43352	0.52387
Ensemble: union	0.43553	0.52739
Full pipeline, not submitted	**0.45777**	0.46914

Selected 3rd place · Only public F1 was visible at selection time.

Remove …	Private	Δ
Full pipeline	0.45777	–
kNN blend	0.44521	–0.013
habitat-fit demotion	**0.40446**	**–0.041**
temporal fusion	0.40079	–0.004
multi-scale → single 5×5	0.38492	–0.016
ViT-L → ViT-B (same scales)	0.36472	–0.036

Cumulative ablation, top to bottom.

The runs that looked best on the **11% public** split generalised worst. Our strongest configuration scored 0.469 public and was never selected.

What did not work

three training-side and augmentation ideas

Noisy-student on LUCAS

Teacher pseudo-labels 212k unlabelled LUCAS field photos; a cross-region transformer student learns from them.

No gain. LUCAS photos and species do not match the eight test sites, so the student learns the wrong distribution.

TileQ-Decoder

Label-as-query transformer (ML-Decoder style) over the 16 tile embeddings, trained on synthetic pseudo-quadrats built in embedding space.

0.12 F1 vs 0.31 for plain max-pool. Synthetic supervision could not be anchored to test-like tiles.

SAM 3 instance crops

Segment plants with text prompts, add each crop to the tile pool alongside the grid tiles.

No consistent gain. Irregular crops sit off the distribution the classifier and index were calibrated on.

Common thread: anything that shifted the distribution between supervision and test lost to aggregation over cached in-domain probabilities.

Takeaways

- **Dataset structure beat model complexity.** Where, at what altitude, and how often a plot was revisited gave the largest gains, for free at inference time.
- **Inference-time beat training-time.** Everything that worked ran on cached per-tile probabilities. Everything that retrained lost.
- **The public leaderboard is small enough to overfit.** Our best run was never submitted because it looked worse on 11% of the data.

Code, configs, habitat-fit scripts:
github.com/dsgt-arc/plantclef-2026

Thank you.

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